Functions and Function Notation

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

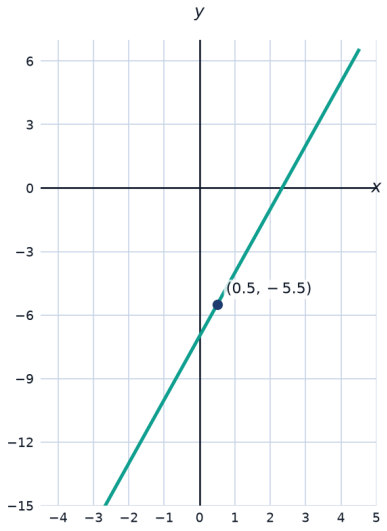
1 [Foundational] • Target: Function notation

Consider the function $f(x) = 2x^2 - 5x + 3$. Evaluate:

a $f(0)$ **b** $f(2)$ **c** $f(-1)$ **d** $f\left(\frac{1}{2}\right)$

2 [Foundational] • Target: Function notation

If $f(x) = 3x - 7$ and $f(a) = 11$, find the value of a.



3 [Foundational] • Target: Identifying functions

Determine whether each equation defines y as a function of x:

a $y = 2x + 1$ **b** $x = y^2$

c $y = |x|$ **d** $x^2 + y^2 = 9$

4 [Foundational] • Target: Identifying functions

Explain how the vertical line test is used to decide whether a graph represents a function.

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5 [Foundational] • Target: Combining functions

A tutoring center charges a one-time registration fee plus an hourly rate, modeled by $C(n) = 250 + 45n$ riyals for n hours of tutoring.

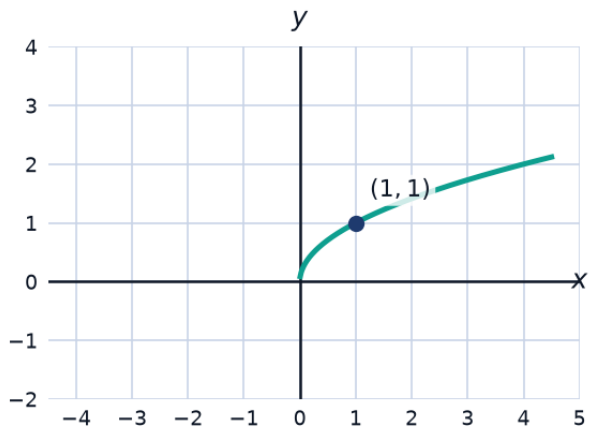
a Interpret the value $C(0)$ in context.

b Find $C(12)$.

c Solve $C(n) = 925$ and interpret your answer.

6 [Foundational] • Target: Combining functions

Let $f(x) = \sqrt{x}$ and $g(x) = x - 4$. Find the domain of $(f \circ g)(x)$.



7 [Foundational] • Target: Piecewise-defined functions

A function is defined by $f(x) = x + 4$ for $x < 1$, and $f(x) = 2x + 1$ for $x \geq 1$. Evaluate:

a $f(-3)$ **b** $f(0)$ **c** $f(1)$ **d** $f(4)$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Piecewise-defined functions

A phone plan charges a flat 60 riyals for up to 5 GB of data, then 15 riyals per additional GB beyond 5 GB.

a Write a piecewise formula $C(x)$ for the monthly cost, where x is data used in GB.

b Find $C(3)$ and $C(8)$.

c Is C continuous at $x = 5$? Explain.

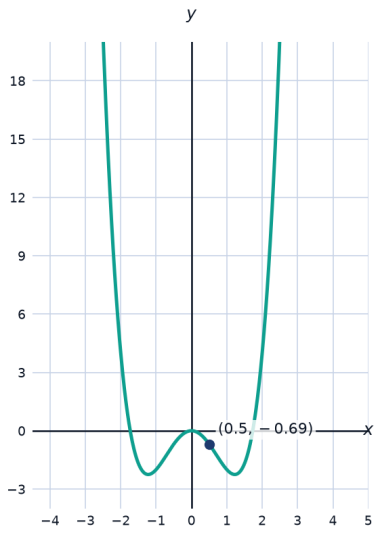
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9 **[Intermediate]** • **Target: Piecewise-defined functions**

Sketch-reasoning: the graph of $f(x) = |x - 2|$ can be written as a piecewise function without absolute value bars. Write it as such, and state $f(-1)$ and $f(5)$.

10 **[Intermediate]** • **Target: Even, odd, and one-to-one functions**

Determine whether each function is even, odd, or neither:



a $f(x) = x^4 - 3x^2$

b $f(x) = x^3 - x$

c $f(x) = x^2 + 2x$

11 **[Intermediate]** • **Target: Even, odd, and one-to-one functions**

Determine whether each function is one-to-one (passes the horizontal line test). If not, give a horizontal line that crosses the graph twice.

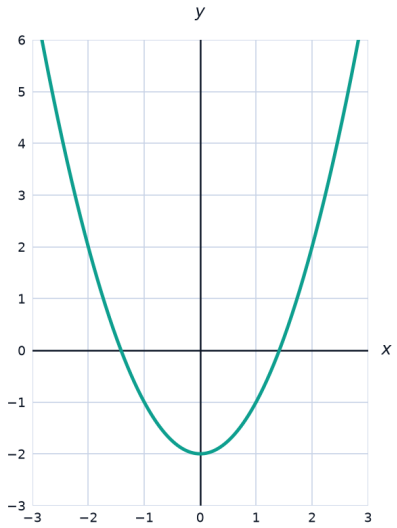
a $f(x) = 2x + 5$

b $f(x) = x^2$

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12 **[Intermediate]** • **Target: Reading function graphs**

The graph of $f(x) = x^2 - 2$ is shown. Use it to estimate $f(1)$, and state the interval(s) where f is increasing.



13 **[Intermediate]** • **Target: Reading function graphs**

Using the graph of $f(x) = x^2 - 2$ above, estimate all x-values where $f(x) = 2$.

14 **[Intermediate]** • **Target: Average rate of change from function notation**

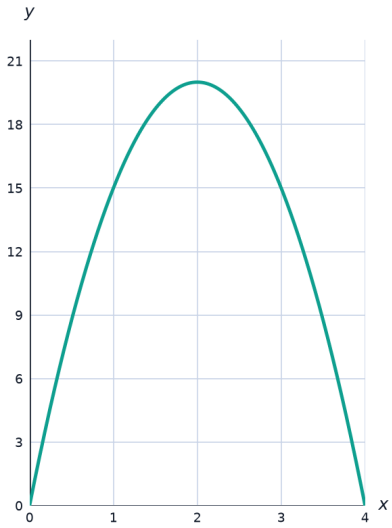
For $f(x) = x^2 + 1$, find the average rate of change of f from $x = 1$ to $x = 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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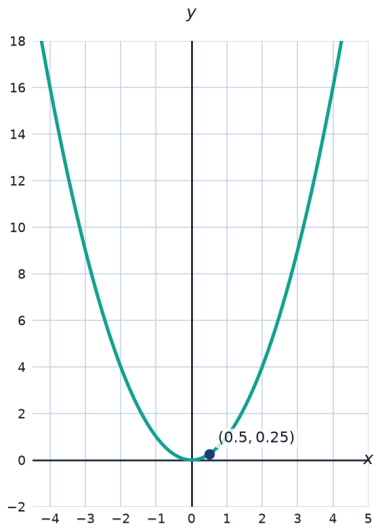
15 [Challenging] • Target: Average rate of change from function notation • Cross-Lesson: 5.02 The Dot Product

A ball's height is modeled by $h(t) = -5t^2 + 20t$ (meters, seconds). Find the average rate of change of height from $t = 1$ to $t = 2$, and interpret the sign of your answer.



16 [Challenging] • Target: Difference quotients • Cross-Lesson: 5.03 Parametric Equations

For $f(x) = x^2$, find and simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$.



17 [Challenging] • Target: Difference quotients • Cross-Lesson: 6.01 Polar Coordinates and Conversions

For $f(x) = 3x + 2$, find and simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$, and explain why the result makes sense for a linear function.

18 [Challenging] • Target: Inverse functions (introduction) • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the inverse of $f(x) = 3x - 4$, and verify by checking $f(f^{-1}(x)) = x$.

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19 [Challenging] • Target: Inverse functions (introduction) • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Explain why $f(x) = x^2$ (with domain all reals) does not have an inverse function, but $f(x) = x^2$ restricted to $x \geq 0$ does.

20 [Challenging] • Target: Identifying functions • Cross-Lesson: 7.02 Logarithms and Their Properties

Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2 + 5x - 3}{x^2 + 4x + 3}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Function notation

A function is defined by $f(x) = x^2$ for $x < 3$ and $f(x) = -2x + 1$ for $x \geq 3$. Find $f(3)$.

a. -6

b. -4

c. 9

d. -5

22 [Foundational] • Target: Identifying functions

Determine whether $f(x) = x^5 + 2x$ is even, odd, or neither.

a. Even

b. Odd

c. Neither

d. Cannot be determined

23 [Foundational] • Target: Combining functions

A piecewise function is defined by $f(x) = 2x + 3$ for $x < 1$ and $f(x) = x + k$ for $x \geq 1$. Find k so that f is continuous at $x = 1$.

a. 5

b. 6

c. 4

d. 1

24 [Foundational] • Target: Piecewise-defined functions

Classify $f(x) = x^2$ as even, odd, or neither.

a. Even

b. Neither

c. Odd

d. Cannot be determined

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25 [Foundational] • Target: Even, odd, and one-to-one functions

The function f is defined by $f(x) = x^2 - 6x + 13 \quad \text{for } x \geq 3$. Determine the complete set of real values of the constant k for which the equation $f^{-1}(x) = |2x - k|$ has exactly one real solution.

a. $k = \frac{39}{8}$ or $5 < k < 11$

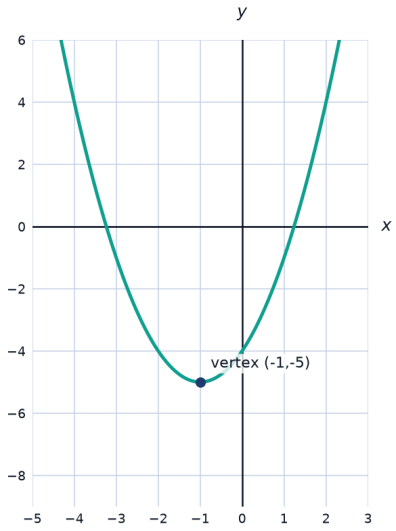
b. $k = \frac{39}{8}$ or $5 \leq k < 11$

c. $\frac{39}{8} < k \leq 5$ or $k \geq 11$

d. $k = \frac{39}{8}$ only

26 [Foundational] • Target: Reading function graphs

The graph of $g(x) = f(x + 1) - 5$ is obtained from $f(x)$ by:



a. Shifting 1 units left and 5 units down

b. Shifting 1 units right and 5 units down

c. Shifting 1 units right and 5 units up

d. Reflecting over the x-axis

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27 [Foundational] • Target: Average rate of change from function notation

Consider the function $f(x) = \frac{x^3 - 8}{x^2 - 4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$?

a. Limit at a is 3, $\lim_{x \to b^-} f(x) = -\infty$

b. Limit at a is 3, $\lim_{x \to b^-} f(x) = \infty$

c. Limit at a is 1, $\lim_{x \to b^-} f(x) = -\infty$

d. Limit at a is 1, $\lim_{x \to b^-} f(x) = \infty$

28 [Foundational] • Target: Difference quotients

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of a.

a. 1

b. 2

c. 3

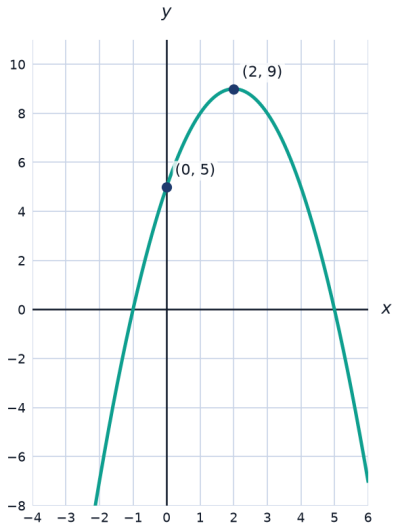
d. 4

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Inverse functions (introduction)**

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the xy-plane with vertex $(2, 9)$ and y-intercept $(0, 5)$. A linear function is defined by $g(x) = mx + 1$. If the graph of g passes through the vertex of the graph of f, what is the other x-coordinate where the graphs of f and g intersect?



a. -4

b. -2

c. -1

d. 1

30 **[Intermediate]** • **Target: Function notation**

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

a. a = 5, b = 3

b. a = -5, b = 3

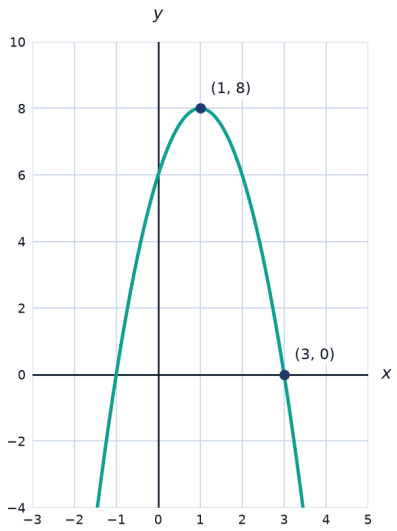
c. a = 5, b = 2

d. a = -5, b = 2

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31 **[Intermediate]** • **Target: Identifying functions**

The graph of the quadratic function $f(x)$ is shown in the figure, with its vertex at $(1, 8)$ and an x-intercept at $(3, 0)$. A second function g is defined by $g(x) = f(x - 2) + k$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, what is the value of k?



a. -10

b. -6

c. 6

d. 10

32 **[Intermediate]** • **Target: Combining functions**

Consider the rational function $f(x) = \frac{ax^2 + bx + c}{dx^2 + ex + f}$, where a, b, c, d, e, f are non-zero constants. Given that $\lim_{x \to \infty} f(x) = \frac{1}{2}$ and $\lim_{x \to -1} f(x) = \infty$, determine the value of $\frac{a}{d}$.

a. $\frac{1}{2}$

b. 1

c. 2

d. Cannot be determined

33 **[Intermediate]** • **Target: Piecewise-defined functions**

Consider the rational function $f(x) = \frac{x^2 - 9}{x - 3}$. What is the value of $f(x)$ as x approaches 3?

a. 0

b. 6

c. Undefined

d. 3

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34 [Intermediate] • Target: Even, odd, and one-to-one functions

Consider the rational function $f(x) = \frac{x^2 - 4}{x - 2}$. What is the value of the function as x approaches 2?

a. 4

b. 0

c. Undefined

d. 2

35 [Intermediate] • Target: Reading function graphs

For $0 < x < 1$, simplify $\cos\left(\sin^{-1}(x)\right)$ as an algebraic expression in x.

a. $\sqrt{1 - x^2}$

b. $\frac{1}{\sqrt{1 - x^2}}$

c. $1 - x^2$

d. $\sqrt{x^2 - 1}$

36 [Intermediate] • Target: Average rate of change from function notation

What is the domain and range of $y = \tan^{-1}(x)$?

a. Domain $(-\infty, \infty)$, range $(-\infty, \infty)$

b. Domain $(-\infty, \infty)$, range $[0, \pi]$

c. Domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, range $(-\infty, \infty)$

d. Domain $(-\infty, \infty)$, range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

37 [Intermediate] • Target: Difference quotients

What is the domain and range of $y = \sin^{-1}(x)$? (variant 5)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, range $[-1, 1]$

c. Domain $[-1, 1]$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

d. Domain $(-\infty, \infty)$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Inverse functions (introduction) • Cross-Lesson: 6.02 Polar Equations and Graphs

Consider the rational function $f(x) = \frac{x^2 - 9}{x - 3}$. What is the value of $f(4)$?

a. 7

b. 1

c. 5

d. 13

39 [Challenging] • Target: Function notation • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

A 13 m ladder leans against a vertical wall and reaches 12 m up the wall. Find the angle the ladder makes with the ground, to the nearest tenth of a degree.

a. 42.7°

b. 22.6°

c. 72.4°

d. 67.4°

40 [Challenging] • Target: Identifying functions • Cross-Lesson: 7.02 Logarithms and Their Properties

A kite string 50 m long is pulled taut, and the kite is 32 m above the hand holding it. Find the angle of elevation of the string, to the nearest tenth of a degree.

a. 50.2°

b. 39.8°

c. 32.6°

d. 44.8°

41 [Challenging] • Target: Combining functions • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

A 15 m rope is tied to the top of a pole and staked to level ground 9 m from the base of the pole. Find the angle the rope makes with the ground, to the nearest tenth of a degree.

a. 36.9°

b. 31.0°

c. 58.1°

d. 53.1°

42 [Challenging] • Target: Piecewise-defined functions • Cross-Lesson: 7.04 Applications: Growth and Decay

A ramp rises 3 m over a horizontal distance of 20 m. Find its angle of inclination, to the nearest tenth of a degree.

a. 8.5°

b. 81.5°

c. 13.5°

d. 17.1°

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43 [Challenging] • Target: Even, odd, and one-to-one functions • Cross-Lesson: 8.01 Rational Functions and Asymptotes

A vertical pole 6 m tall casts a shadow 9 m long. Find the angle of elevation of the sun, to the nearest tenth of a degree.

a. 33.7°

b. 56.3°

c. 67.4°

d. 38.7°

44 [Challenging] • Target: Reading function graphs • Cross-Lesson: 8.02 Graphing Rational Functions

For what value of k is $f(x) = -2x + k$ (for $x < 5$) and $f(x) = x^2$ (for $x \geq 5$) continuous at $x = 5$?

a. 36

b. 34

c. -35

d. 35

45 [Challenging] • Target: Average rate of change from function notation • Cross-Lesson: 8.03 Introduction to Limits

If $f(x) = \sqrt{x - 5}$ and $g(x) = x^2 - 4$, what is the domain of $f(g(x))$?

a. $-3 \leq x \leq 3$

b. $x \leq -3$ or $x \geq 3$

c. $x \geq 3$

d. All real numbers

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